

04.02 Fractional Exponents

Concept summary and key formulas

Key ideas

A fractional exponent combines powers and roots.

The denominator gives the root; the numerator gives the power.

Rewrite between radical and exponential notation before simplifying.

Key formulas

$$\begin{aligned}a^{1/n} &= \sqrt[n]{a} \\a^{m/n} &= \sqrt[n]{a^m} \\a^p a^q &= a^{p+q}\end{aligned}$$